

Catalytic Ladder Propagation

Federated Retrieval of Heterogenous Sources Catalytic Data through Translated Graph Verdicts

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Abstract

A knowledge graph is only as useful as the person who built it. To ask a question of one, you must know how it was modelled and how to express your question in that model; both requirements bind, and together they bound the work that can be done with the artefact. This paper describes an architecture in which neither requirement binds, and demonstrates it on twenty-eight questions — twelve supplied by practitioners without reference to this framework, sixteen added to exercise outcomes the twelve do not reach.

The architecture has four stages. Records are *fetched* from heterogeneous sources through a plan language whose terms denote result sets rather than graph patterns, so that portability across backends is structural rather than an integration effort. Records are *translated* into weighted contact graphs carrying a distinguished medium vertex. A *ladder* — a sequence of contacts, each carrying one number in $[0, 1]$ — is applied to the translated graph as a stencil. The output is a *verdict* drawn from six values, of which an answer set is one.

We derive the substrate rather than assuming it. From a single axiom about bounded phase space we obtain oscillatory necessity, a partition coordinate system, a three-coordinate entropy space, and from these the contact graph and its *floor* $\beta > 0$: an irreducible cost of individuating anything. Identity is then a minimum cut, never a label. The ladder follows, together with its composition law $\pi(L) = 1 - \prod_i (1 - \pi_i)$.

Four results are the point. First, *retrieval cannot express admissibility*: we exhibit two systems with identical triples for which no retrieval query

differs and the propagation verdict differs, and we execute the separation on two independently developed SPARQL engines over thirteen query forms, with a control confirming the battery is not blind (Theorem 6.4, Section 10.2). Second, *solvent role and reaction direction are computed rather than annotated*: an ordered active-site water and a bulk water of identical chemical identity receive different roles, and one reaction identifier receives three different direction verdicts in three media (Theorem 7.4, Theorem 7.9). Third, we find and prove a *saturation asymmetry* we did not anticipate: flooding a product cannot reverse a chain, because the attainable bias saturates at $-\log(1 + \tau/\mu_0)$, while depleting a reactant is unbounded (Theorem 7.11). Fourth, the architecture *refuses*: seven of the twenty-eight questions are declined, each with a named blocker and a statement of what would unblock it.

We replicate, on **3 September 2026**, a divergence between two SPARQL spellings that the specification defines as one abstract query: against a public reaction endpoint they return 2 and 397, while two independent local engines agree with each other and with hand computation on a miniature of the same data shape. We report this as a coverage measurement rather than as a defect, and state the condition under which that reclassification holds.

Fifty-eight checks are reported across four experiments, of which four are negative controls that must fail on well-formed input and one is recorded non-discriminating and excluded from the score. Every quantitative claim is computed by the accompanying suite or is marked as not established.

Keywords: federated retrieval; catalysis data; contact graph; individuation floor; verdict; refusal; solvent role; reaction directionality; knowledge graph incompleteness; negative control.

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Part I

The Problem

1 Two requirements that bind

1.1 The modelling requirement and the querying requirement

Consider what must be true before a biochemical knowledge graph answers a question.

Someone must have decided that the entities in the question are entities in the model, that the relation asked about is a relation in the schema, and that the particular instance is worth recording. That is the *modelling requirement*, and it is discharged by a curator, in advance, without knowing which questions will be asked.

Then the person asking must know what that curator decided. They must know which class the entity was filed under, which property carries the relation, and which of several plausible

spellings the store will match. That is the *querying requirement*, and it is discharged by the asker, at query time, by reading documentation or by knowing the curator.

Both requirements bind, and their conjunction is the limit on what the artefact can do. A question outside the model cannot be asked. A question inside the model but outside the asker’s knowledge of the model will not be asked, or will be asked wrongly and answered wrongly with no indication that anything went amiss. The artefact is therefore only as useful as the individual who generated it, and only as useful again as the individual querying it.

This paper is about removing both requirements for a specific and important class of question. We do not claim to remove them in general.

1.2 The failure that motivates the architecture

The failure mode is worth stating concretely, because it is invisible.

Ask a system: *which reactions have exactly four participants?* Suppose the corpus contains three such reactions, and the system returns the empty set. There are at least five distinct situations that produce that output.

1. The model has no cardinality constructs, so the question was never representable.
2. The model has them, but under an open-world reading the count is not entailed: nothing rules out a fourth participant that was merely not recorded.
3. The corpus genuinely contains no such reaction.
4. The compiler silently dropped the cardinality restriction while lowering the question, so a well-formed query ran and matched nothing.
5. The reasoner exhausted a budget and returned partial results without saying so.

These five warrant five different actions from whoever receives the answer, and one output is produced for all of them. The output is not wrong in any way inspection reveals: it is well formed, correctly typed, promptly returned, and in case (2) it is even *correct* under the semantics

the system nominally implements. It is uninformative in a way that is invisible.

An architecture in which those five produce five different outputs is not a reporting improvement bolted onto a query engine. It is a different kind of object: the system becomes an instrument, and its output at each question is a verdict, of which an answer set is one possible constituent.

1.3 Why incompleteness is the expected state

We do not treat the incompleteness of a catalysis knowledge graph as a temporary condition to be engineered away. Three independent sources guarantee it.

The first is expressive. Every decidable fragment of first-order logic omits constructions that some question will need, and the trade-off between expressive power and tractability of reasoning is a theorem-level fact about the fragments rather than an engineering artefact (Baader et al., 2003; Motik et al., 2009a). A model chosen so that reasoning terminates is a model in which some questions are not statable.

The second is empirical. Curated corpora are assembled by finite human effort from an open and growing literature. Reaction databases record what has been curated, not what exists (Alcántara et al., 2012; Bansal et al., 2022), and chemical ontologies partition molecules along axes chosen for chemical rather than process convenience (Hastings et al., 2016). Absence in a corpus is not evidence of absence in the world. The FAIR programme (Wilkinson et al., 2016) is in substantial part a response to this: if knowledge is stored, then its findability and reuse are properties of the storage, each separately engineered and separately maintained (Gaudet and Dessimoz, 2017).

The third is computational. Even where a question is statable and entailed, the reasoner may not finish; description-logic reasoning is intractable in the worst case for expressive profiles (Glimm et al., 2014; Motik et al., 2009b).

These three are independent: a question may be inexpressible in a model whose corpus is rich and whose engine is fast; expressible and supported but unentailed; or expressible, entailed and beyond budget.

1.4 What this paper does

- (i) Derives the substrate — contact graph, floor, cut, ladder — from one axiom about bounded phase space, so that no primitive is posited (Sections 2, 4 and 5).
- (ii) Proves that retrieval over stored triples cannot express admissibility, and *executes* the separation on two SPARQL engines (Theorem 6.4, Section 10.2).
- (iii) Gives the medium its content, from which solvent role and reaction direction become computations rather than annotations (Section 7).
- (iv) Specifies the plan language and the six verdicts, and proves that a verdict other than **answer** may never carry a payload (Section 8).
- (v) Resolves twenty-eight questions, answering twenty-one and refusing seven with named blockers (Section 10.3).
- (vi) Reports what the framework does not do (Section 11).

1.5 What we do not claim

We do not claim the architecture is more accurate than curation; it computes different quantities. We do not claim that any number here has been validated against a laboratory measurement — none has, and Section 11 says so. We do not claim that knowledge graphs are useless: Theorem 6.7 shows that the signature view and the chain view answer *disjoint* question sets, so the correct conclusion is that both are needed and neither subsumes the other. And we do not claim that the endpoint divergence of Section 10.4 is a defect; we establish that it exists and decline to diagnose a third party’s service.

Part II

Deriving the Substrate

2 One axiom

Everything below follows from a single statement about what it takes to persist.

Axiom 2.1 (Bounded phase space). Every persistent dynamical system occupies a bounded region Ω of phase space with finite measure $\mu(\Omega) < \infty$, and that region admits a hierarchical partition into distinguishable cells.

The axiom is empirically motivated rather than metaphysical. A molecule in solution is confined by its solvation shell; an electron is bound by a Coulomb potential; a vibrational mode is bounded by a potential well. Systems that are not bounded do not persist — they disperse.

Proposition 2.2 (Persistence requires boundedness). *In unbounded phase space, generic configurations are not persistent.*

Proof. For a region R of finite measure inside Ω with $\mu(\Omega) = \infty$, the long-run fraction of time an ergodic trajectory spends in R is $\mu(R)/\mu(\Omega) = 0$. Trajectories spend zero fraction of time in any bounded region, so no bounded configuration persists. $\square$

Theorem 2.3 (Oscillatory necessity). *For a measure-preserving system on a bounded region, oscillatory dynamics — motion returning to neighbourhoods of previous states — is the only mode consistent with Theorem 2.1.*

Proof. By the Poincaré recurrence theorem (Poincaré, 1890; Walters, 1982), for a measure-preserving flow on a finite-measure space, almost every point of any positive-measure set returns to that set infinitely often, with mean return time inversely proportional to the measure (Kac, 1947). We eliminate the alternatives. Static dynamics distinguishes no states and so supports no individuation. Monotone dynamics admits a function strictly increasing along trajectories, so some positive-measure set is never revisited, contradicting recurrence. Chaotic dynamics with positive Lyapunov exponent destroys the reproducibility that categorical labelling requires, since the same initial condition yields different labels. Only recurrent — oscillatory — motion remains. $\square$

Corollary 2.4 (Discrete modes). *Oscillation on a bounded domain decomposes into countably many discrete modes: the Koopman operator on such a domain has a point spectrum (Koopman, 1931), so any observable expands as $f = \sum_k c_k e^{i\omega_k t}$.*

Remark 2.5 (What the axiom buys, and what it does not). Theorem 2.1 yields the *form* of the substrate: that individuation is discrete, that it costs something, and that the cost has a floor. It does not yield any numerical value. Every number in this paper comes from a measurement, a fixture, or an explicitly illustrative choice, and we mark which.

3 Partition coordinates and entropy space

Definition 3.1 (Partition, refinement). A *partition* of Ω is a family of disjoint non-empty subsets covering it. A partition $\mathcal{Q}$ *refines* $\mathcal{P}$ if every cell of $\mathcal{Q}$ lies in a cell of $\mathcal{P}$. Theorem 2.1 supplies a hierarchical sequence $\mathcal{P}_0 \prec \mathcal{P}_1 \prec \dots$.

Proposition 3.2 (Finite capacity forces categories). *An observer with finite information capacity I must approximate a continuous state by the cell containing it: the finest partition available has at most 2^I cells.*

Proof. Specifying a point of a continuum to arbitrary precision requires unbounded information. With $I < \infty$ bits the observer distinguishes at most 2^I alternatives, so its description is a cell assignment. $\square$

This is the origin of the entropy coordinates we use to weight contacts. A biomolecular object at a given resolution is described by three classes of observable, and no fourth class enters its equilibrium behaviour.

Definition 3.3 (Entropy coordinates). For an object X define

$$S_k(X) = \frac{\ln \Omega_{\text{cfg}}(X)}{\ln \Omega_{\text{max}}} \quad \text{configurational: how many arrangements}$$

$$S_t(X) = \frac{\sum_r m_r \|\mathbf{x}_r - \bar{\mathbf{x}}\|^2}{\mathcal{N}_T} \quad \text{spatial: how mass is distributed}$$

$$S_e(X) = \frac{-\sum_i p_i \ln p_i}{\ln N_{\text{vib}}} \quad \text{energetic: how states spread in energy along a cut while distances do not.}$$

with p_i the Boltzmann weights; S_e is a normalised Shannon entropy (Cover and Thomas, 2006; Shannon, 1948). Write $\mathbf{s}(X) = (S_k, S_t, S_e) \in \mathcal{S} := [0, 1]^3$.

Proposition 3.4 (Independence and sufficiency). *The three coordinates are pairwise functionally independent, and together they determine equilibrium thermodynamic behaviour to the resolution of the microstate enumeration.*

Proof. Independence. Two conformers of one molecule share a rotamer count (equal S_k) but differ in mass distribution (S_t). One molecule in two solvents of different dielectric shares its conformational enumeration but differs in energy spectrum (S_e). Two isomers may share a radius of gyration while differing in spectrum, since the spectrum depends on bonding topology and not on geometry alone.

Sufficiency. The canonical partition function $Z = \sum_i g_i e^{-E_i/k_B T}$ determines every equilibrium quantity. The multiset $\{(E_i, g_i)\}$ is what S_e records together with the arrangement count S_k ; S_t fixes the spatial embedding deciding which arrangements are sterically realisable. No fourth class of observable enters Z . $\square$

Remark 3.5 (The coordinates carry no free parameter). None of Theorem 3.3 contains a fitted constant. Ω_{cfg} is a rotamer or conformer count; the numerator of S_t is a mass-weighted second moment computable from any structure; the p_i come from a vibrational spectrum. The normalisers are fixed once for a reference set and are not per-object.

4 Contact, cut, and the floor

Definition 4.1 (Contact graph). A *contact graph* is a finite weighted graph $\Gamma = (V, E, w)$ with $w : E \rightarrow (0, \infty)$, carrying a distinguished medium vertex $\mathbf{m} \in V$ to which every other vertex is connected. We write $w(F) = \sum_{e \in F} w(e)$ and $\Omega = w(E)$.

The weight $w(\{u, v\})$ is the *cost of separating u from v* , never a distance between them. The distinction matters because separation costs add along a cut while distances do not. The medium is what an item is separated *from*: in solution, the solvent; in a reactor, the bulk phase. It is a structural slot, not a chemical assumption.

Definition 4.2 (Cut and separation cost). For $S \subseteq V$ the *cut* $\partial(S)$ is the set of edges with

exactly one endpoint in S . For $v \neq \mathfrak{m}$ the *separation cost* is

$$\sigma(v) = \min_{S \ni v, \mathfrak{m} \notin S} w(\partial(S)). \quad (4)$$

Axiom 4.3 (Non-completeness). There is no assignment making every item a medium for every other: no family $\{S_v\}$ with $v \in S_v$, $\mathfrak{m} \notin S_v$ and $w(\partial(S_v)) = 0$ for all v .

This is an ordering claim about the world. If every item could serve as medium for every other, no item would be distinguished by what it is separated from, and “being in contact” would carry no information. Every observed molecular system violates it: a buried residue is not in contact with bulk solvent the way a surface residue is.

Theorem 4.4 (The contact floor). *Under Theorem 4.3, $\sigma(v) > 0$ for every $v \neq \mathfrak{m}$, and*

$$\beta := \min_{v \neq \mathfrak{m}} \sigma(v) > 0. \quad (5)$$

Proof. Suppose $\sigma(u) = 0$ for some u . The minimum in Equation (4) is over a finite non-empty family, so it is attained at some S_u with $w(\partial(S_u)) = 0$. Since $w > 0$ on every edge this forces $\partial(S_u) = \emptyset$, so u and $\mathfrak{m}$ lie in different components, contradicting the connectivity of Theorem 4.1. Hence every $\sigma(v) > 0$, and V is finite, so the minimum of finitely many positive numbers is positive. $\square$

Remark 4.5 (The floor is not an edge weight). β is a minimum over vertices of a minimum over subsets. It is not the smallest edge weight, and the difference is not pedantic: a graph may contain an edge of weight 10^{-12} and still have $\beta \approx 1$, because a cheap edge inside an expensive cut does not make the cut cheap. Our suite checks exactly this case (Section 10.1), and Figure 1(d) shows two graphs with equal minimum edge weight and different floors.

Theorem 4.6 (Uniformity requires a resolution bound). *Positivity of each $\sigma(v)$ does not imply a positive infimum over an unbounded family of refinements. A uniform floor exists exactly when the system is resolution-bounded: the individualisations it can perform have a finest member.*

Proof. The numbers $1/n$ are all positive with infimum zero (Rudin, 1976), so positivity of each

member of a family does not give positivity of the infimum. If the family of attainable separation costs has a least element $\mu_{\min}$, then $\beta = \mu_{\min} > 0$ by Theorem 4.4. Physical systems are resolution-bounded because the number of distinguishable cells in a bounded region of phase space is finite, by Theorem 2.1. $\square$

We flag this dependence because every result downstream inherits it, and Figure 1(b) exhibits both branches.

Proposition 4.7 (Identity is a region, not a point). *There are contact graphs in which every minimiser of Equation (4) for some v contains more than one vertex. The identity of v is then a property of a region containing v , not of v alone.*

Proof. Let $A = \{a_0, \dots, a_{n-1}\}$ be a clique with internal weight W , joined to the rest of the graph by a single edge of weight 1 at a_0 , with the medium adjacent only to the remainder. Any $S \ni a_0$ omitting some a_j must cut an internal edge, costing at least W ; taking $S = A$ costs 1. For $W > 1$ every minimiser contains all of A . $\square$

Remark 4.8. Read chemically: a residue in a tightly packed core is a_0 . Severing it individually costs more than severing the whole core from solvent, so its identity — what it is separated from, and at what cost — is a property of the core. Any scheme assigning identity residue-by-residue is assigning something other than identity. Our suite measures this (Section 10.1).

Theorem 4.9 (Labels carry no invariant). *Let $\lambda : V \rightarrow L$ be a labelling into a set with no structure derived from Γ , and let $\text{Aut}_{\mathfrak{m}}(\Gamma)$ be the automorphism group fixing $\mathfrak{m}$. Then λ is invariant only if constant on orbits; and if λ is injective on an orbit of size > 1 , it carries information not determined by $(V, E, w, \mathfrak{m})$.*

Proof. Any property definable from $(V, E, w, \mathfrak{m})$ is preserved by every $\phi \in \text{Aut}_{\mathfrak{m}}(\Gamma)$, so it cannot distinguish v from $\phi(v)$. If λ separates two vertices in one orbit while every Γ -definable property agrees on them, the distinction λ draws is not Γ -determined. $\square$

Corollary 4.10 (Why cross-references must be curated). *Two resources labelling the same system with independent labellings cannot compute their correspondence from the labels; any*

mapping between them is an assertion, not a derivation, and must be maintained externally.

Theorem 4.10 is the formal content of an observation familiar to anyone who has maintained an identifier mapping: the decay is not a failure of discipline but a consequence of having located identity in the label (McMurry et al., 2017; Wren, 2008). Locate it in the cut instead and the correspondence becomes computable, by Equation (4).

5 The ladder

We now define the object applied as a stencil.

Definition 5.1 (Alignment). For disjoint regions x, x^* the *alignment* is $a(x, x^*) = \sigma(x, x^*)/\Omega \in (0, 1]$, where $\sigma(x, x^*)$ is the minimum weight of a cut separating them.

Lemma 5.2 (Alignment is floored). $a(x, x^*) \geq \beta/\Omega > 0$.

Proof. Any separating cut is non-empty, and by Theorem 4.4 its weight is at least β ; divide by $\Omega > 0$. The upper bound is $\partial(S) \subseteq E$. $\square$

Definition 5.3 (Rung and power). A *rung* toward target x^* is a map R on regions whose application commits a contact. Its *power* at x is the fraction of the above-floor alignment it closes,

$$\pi(R, x) = \frac{a(x, x^*) - a(R(x), x^*)}{a(x, x^*) - \beta/\Omega} \in [0, 1]. \quad (6)$$

A rung is *inert* if $\pi = 0$ and *absolute* if $\pi = 1$.

Definition 5.4 (Ladder). A *ladder* is a finite sequence of rungs applied in order, each to the output of the last, with *power sequence* $(\pi_1, \dots, \pi_n)$.

Remark 5.5 (What a rung is not). Theorem 5.3 mentions no property of R beyond the alignment it closes: not what it is made of, not how it operates, not how long it takes. Two rungs with the same power are, at this resolution, the same rung. This is what makes the ladder a stencil: it can be laid over records fetched from any source, because it asks of them only a number. It is worth contrasting with the usual criteria for whether two catalysts are “the same kind” — mechanism, fold, active-site residues, sequence homology — which are known to disagree with one another (Khersonsky and Tawfik, 2010).

Theorem 5.6 (Multiplicative composition). A ladder with power sequence $(\pi_1, \dots, \pi_n)$ has *composite power*

$$\pi(L) = 1 - \prod_{i=1}^n (1 - \pi_i). \quad (7)$$

Proof. Write $g_i = a(x_i, x^*) - \beta/\Omega$ for the above-floor gap after i rungs. By Equation (6), $\pi_i = (g_{i-1} - g_i)/g_{i-1}$, so $g_i = (1 - \pi_i)g_{i-1}$, and by induction $g_n = g_0 \prod_{i \leq n} (1 - \pi_i)$. The fraction of the initial gap closed is $1 - g_n/g_0$, which is Equation (7). $\square$

Corollary 5.7 (Repetition saturates). *Repeating one rung of power π gives composite $1 - (1 - \pi)^n$, which is strictly increasing, tends to 1, and attains it at no finite n . Copies of what already works have geometrically diminishing returns: the n -th repetition contributes $\pi(1 - \pi)^{n-1}$.*

Proposition 5.8 (Cost of a target). *Reaching composite power π^* with rungs of power at most $\pi_{\max}$ requires at least $\lceil \log(1 - \pi^*) / \log(1 - \pi_{\max}) \rceil$ rungs.*

5.1 Sensitivity, and a correction

Proposition 5.9 (Additive sensitivity). *With $P = \prod_i (1 - \pi_i)$, $\partial\pi(L)/\partial\pi_j = P/(1 - \pi_j)$, which is increasing in π_j and hence maximised at the highest-power rung.*

Proof. Differentiate Equation (7). $\square$

This is a counter-intuitive result: it says the rung worth improving is the strongest, not the bottleneck. We state the qualification it requires, because without it the claim misleads.

Proposition 5.10 (Proportional sensitivity is flat). *Let improvement be proportional to a rung’s remaining headroom, so rung j is raised by $\delta(1 - \pi_j)$ rather than by δ . Then the gain in composite power is δP , independent of j .*

Proof. Multiply the derivative of Theorem 5.9 by $\delta(1 - \pi_j)$; the factors $(1 - \pi_j)$ cancel. $\square$

Remark 5.11 (Which parametrisation applies is an empirical question). The derivative of Theorem 5.9 prices an *additive* increment applied equally at every rung. Under proportional improvement — raising a rung by a fixed fraction of its own headroom, which is what “improve

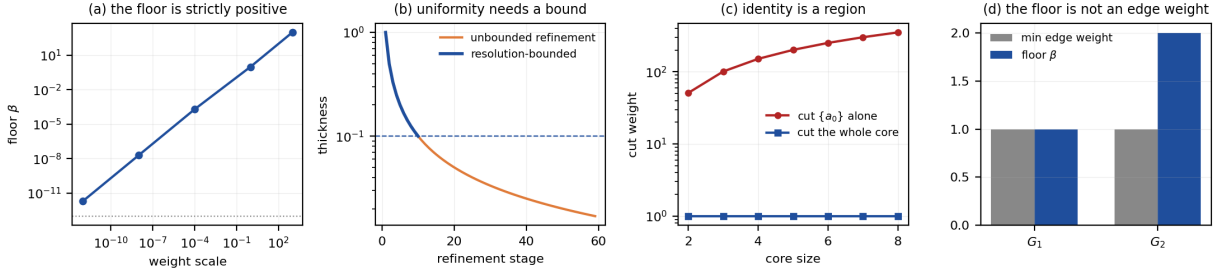


Figure 1: The floor is computed, the uniformity hypothesis does real work, and identity is a region rather than a point. (A) Contact floor β computed—not assumed—over instances spanning twelve orders of magnitude of weight scale, including an adversarial case whose smallest edge weight is 10^{-12} . Every instance returns a strictly positive floor. The adversarial instance is the informative one: its floor is $1 + 10^{-12}$ and *not* 10^{-12} , because that light edge lies inside every cut that would use it, and a cheap edge inside an expensive cut does not make the cut cheap. Positivity follows from finiteness and strict positivity of the weights, not from their scale. (B) The failure branch, drawn so the hypothesis is visible rather than tacit. A system permitted to refine without limit generates thicknesses $1/n$ (orange): each is strictly positive while the infimum is zero, and every ε in a ladder down to 2.5×10^{-3} is eventually cleared. Truncating the same sequence to a resolution-bounded system (blue, first ten stages) restores a positive floor at 0.1 (dashed). Theorem 4.4 is unconditional; the uniform statement of Theorem 4.6 is not, and every result downstream inherits the condition. (C) Identity is a region and not a point (Proposition 4.7). For a vertex buried in a clique of internal weight $W = 50$, severing it alone from the core costs $W(n - 1) + 1$ and grows with core size (red), while severing the whole core from its surroundings costs 1 regardless (blue). Exhaustive enumeration confirms every minimiser contains all four core vertices. Read chemically: the identity of a residue in a packed core—what it is separated from, and at what cost—is a property of the core, so any scheme assigning identity residue-by-residue is assigning something other than identity. (D) The floor is not an edge weight, and the difference is what Theorem 6.4 turns on. Two contact graphs with *identical* minimum edge weight 1.0 (grey) have floors 1 and 2 (blue). A query over stored weights sees the grey bars; the accountability verdict depends on the blue ones.

this catalyst by ten per cent” ordinarily means — the sensitivity is flat. Our suite measures both: the additive argmax is the highest-power rung in 5000/5000 ladders, and the proportional spread across rungs is 1.1×10^{-16} , machine zero, in 5000/5000 (Section 10.1, Figure 2(d)).

So the counter-intuitive direction is real under one parametrisation and absent under another, and which applies depends on how improvements are actually purchased in the laboratory. A check comparing the analytic derivative against a numerical one cannot detect this, because both quantities are the same additive derivative. We state the qualification rather than the headline.

5.2 Closed ladders

A linear ladder runs toward a target. A ring has none: it returns. Composite power is undefined for it, and two invariants take its place.

Definition 5.12 (Circulation and uniformity). For a closed ladder with profile $(\pi_1, \dots, \pi_n)$, all $\pi_i < 1$,

$$\varrho = -\sum_{i=1}^n \log(1-\pi_i), \quad v = \max\left(0, 1 - \frac{\text{sd}(\pi)}{\text{mean}(\pi)}\right) \quad (8)$$

Proposition 5.13 (Properties). *Both invariants are unchanged by rotation of the cycle; ϱ is additive over concatenation of arcs and vanishes exactly when every rung is inert.*

Proof. Both are functions of the multiset $\{\pi_i\}$, which rotation permutes trivially. Additivity is additivity of a sum. Each term $-\log(1-\pi_i) \geq 0$ with equality iff $\pi_i = 0$. $\square$

Remark 5.14 (Where circulation is redundant, stated plainly). For a *linear* ladder $\varrho = -\log(1-\pi(L))$ exactly, verified to 10^{-13} over 500 random ladders (Section 10.1). It is therefore a monotone reparametrisation of composite power and adds no information whatever in the linear case. We say this rather than let a new symbol suggest a new quantity. Its content is entirely that it is defined *without a target*, so it survives the passage to a cycle where composite power does not.

The invariant that genuinely adds information is v : over 4000 pairs of ladders constructed to share a composite power exactly, 2518 differed in v . A description recording only composite power cannot see the difference.

Proposition 5.15 (Length is not recoverable). *Closed ladders of different length may share both invariants, so n must be carried separately. A uniform cycle of length n at power π and one of length $2n$ at π' with $(1-\pi')^2 = 1-\pi$ have equal ϱ and equal $v = 1$.*

Part III

What Retrieval Cannot Do

6 Admissibility is not retrieval

6.1 Two kinds of question

A query language over a stored graph evaluates patterns against triples. In their core fragments SPARQL and its relatives are conjunctive queries with regular path expressions (Angles et al., 2017; Pérez et al., 2009): evaluation finds homomorphisms from a pattern into the data, so everything returned is a substructure of what was stored. That is the correct semantics for a container.

Definition 6.1 (Retrieval query). Let G be a finite set of triples. A *retrieval query* assigns to G a set of tuples computed by pattern matching and path composition on G alone.

Definition 6.2 (Accountable propagation). Fix $\varepsilon \geq 0$. A propagation from x to x^* is ε -accountable if $a(x, x^*) \leq \beta/\Omega + \varepsilon$. Unfolding Theorem 5.1, this is

$$\sigma(x, x^*) \leq \beta + \varepsilon \Omega. \quad (9)$$

Definition 6.3 (Admissibility query). Given Γ , an item v_0 , a target x^* and a tolerance, the *admissibility query* returns whether an accountable propagation exists, together with its alignment.

At $\varepsilon = 0$ the total Ω cancels and the verdict is a comparison of two numbers: a single minimum cut, and the floor. The first is local to the queried pair. The second is a minimum over *every* vertex in the system. The separation below is the observation that these two can be moved independently, and that a triple records neither.

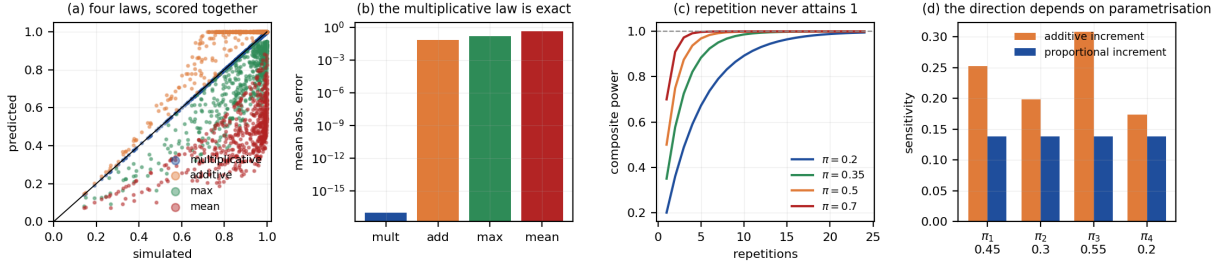


Figure 2: Composition is multiplicative, repetition saturates, and the counter-intuitive direction of control is a property of the parametrisation. (A) Predicted against simulated composite power for four candidate composition laws on randomly generated ladders, scored against a rung-by-rung simulation of the residual gap. The multiplicative law lies on the identity; the additive law saturates and overshoots, the maximum law systematically underestimates, and the mean law fails badly. The comparison is the test: scoring only the favoured law would not have distinguished it from the others. (B) Mean absolute error by law over 4000 ladders, on a logarithmic axis. The multiplicative law is exact to machine precision (0.0); the alternatives err by 5.45×10^{-2} , 1.49×10^{-1} and 4.31×10^{-1} . The gap is not a matter of fit quality but of which law is true. (C) Composite power under repetition of a single rung, for four rung powers. Each curve rises strictly, tends to unity (dashed), and reaches it at no finite n ; the n -th repetition contributes $\pi(1 - \pi)^{n-1}$. Copies of what already works have geometrically diminishing returns, so a ladder of near-duplicate rungs cannot attain a demanding target however long it is made. (D) The correction of Remark 5.11, and the reason this panel exists. Under an *additive* increment applied equally at every rung (orange) the sensitivity $\prod_{i \neq j} (1 - \pi_i)$ varies across rungs and is maximised at the *strongest*, $\pi_3 = 0.55$ —the counter-intuitive direction, reproduced here in 5000/5000 ladders. Under a *proportional* increment, raising a rung by a fixed fraction of its own headroom, the factors cancel and the sensitivity is identical at every rung (blue): measured spread 1.11×10^{-16} , machine zero, in 5000/5000 ladders. Which parametrisation applies is an empirical question about how improvements are purchased, and a check comparing the analytic derivative against a numerical one cannot detect the difference, because both quantities are the same additive derivative.

6.2 The separation

Theorem 6.4 (Retrieval cannot express admissibility). *There exist grounded contact graphs Γ_1, Γ_2 on the same vertex set, with the same contact relation, and a pair (v_0, x^*) , such that*

- (i) $q(G_1) = q(G_2)$ for every retrieval query q , where G_i records the contact relation of Γ_i ; while
- (ii) an accountable propagation from v_0 to x^* exists in one and not in the other, at $\varepsilon = 0$.

Proof. Let $V = \{v_0, x, y_1, \dots, y_n, m\}$ with $n \geq 1$. In both graphs the edges are $\{v_0, x\}$ and $\{u, m\}$ for every $u \neq m$. Both are grounded.

Assign weights. In both, $w(\{v_0, x\}) = 1$ and the medium edges at v_0 and x have weight 1. The graphs differ only in the medium edges at the y_i : weight 1 in Γ_1 , weight 2 in Γ_2 .

(i) The edge sets are equal, so the recorded contact relations coincide as sets of triples. Any q in the sense of Theorem 6.1 is a function of its argument alone, hence agrees on them. A triple asserts that a relation holds and carries no cost, so there is nothing here for retrieval to see.

(ii) The y_i are joined only to m , so they lie on no v_0 – x path and no minimum such cut uses their edges: the cheapest separation places v_0 alone, severing $\{v_0, x\}$ and the medium edge at v_0 , so $\sigma(v_0, x) = 2$ in *both* graphs. The floor is a minimum over all vertices. Isolating v_0 or x costs 2; isolating y_i severs its single medium edge, costing 1 in Γ_1 and 2 in Γ_2 . Hence $\beta_1 = 1$ and $\beta_2 = 2$. Applying Equation (9) at $\varepsilon = 0$: in Γ_2 , $\sigma = 2 \leq 2 = \beta_2$ and the propagation is accountable; in Γ_1 , $\sigma = 2 > 1 = \beta_1$ and it is not. $\square$

Remark 6.5 (What the separation is about). The construction turns on one asymmetry. The queried quantity $\sigma(v_0, x^*)$ depends on weights along v_0 – x^* separations; the threshold β depends on a minimum over every item in the system, including items lying on no path between the two and, from the query’s point of view, entirely irrelevant. Editing a weight at the y_i changes the verdict about (v_0, x^*) without changing anything a pattern-matcher could match.

So *accountability* is a global property expressed in local terms, and a stored graph records only the local terms.

Corollary 6.6 (Attributed triples do not close the gap). *Extending Theorem 6.1 to allow numeric attributes on triples and arithmetic comparison in patterns does not make admissibility expressible.*

Proof. By Theorem 6.4 the verdict depends on β , a minimum over items of a minimum over subsets. Conjunctive queries with regular path expressions are evaluated by matching bounded patterns (Angles et al., 2017; Barceló Baeza, 2013; Pérez et al., 2009); the fragment contains no operator forming a minimum over subsets of the domain, and comparing stored attributes does not introduce one. The obstruction is expressibility, not hardness: a minimum cut is computable in polynomial time (Ford and Fulkerson, 1956; Stoer and Wagner, 1997), so the quantity is cheap to compute and impossible to phrase; multi-terminal variants are equally classical (Gomory and Hu, 1961; Menger, 1927). $\square$

Our suite executes Theorem 6.6 directly: with weights stored as triple attributes, the minimum edge weight is 1.0 in both graphs while the floors are 1 and 2 (Section 10.2).

6.3 Two representations, one system

It would be a misreading of Theorem 6.4 to conclude that stored graphs are useless. The correct conclusion is sharper.

Theorem 6.7 (Representational partition). *A description-logic defined class over participant roles cannot express a residue chain, and a residue chain does not determine the stoichiometric coefficients of its participants. Hence the signature view and the chain view answer disjoint sets of questions, and neither is a refinement of the other.*

Proof. Signatures cannot carry mechanism. A residue chain is a finite sequence with state: the cut at step i is the boundary condition of step $i + 1$. Expressing it requires quantification over positions and a successor relation with transitive closure. *SRQIQ*, the description logic underlying OWL 2, admits transitive roles but not the composition of a transitive role with the value restrictions needed to thread state along it without losing decidability (Baader et al., 2003; Cuenca Grau et al., 2008; Horrocks et al., 2006).

A defined class can therefore state that a reaction *has* a participant with a role, but not that its steps occur in an order; an ordering is exactly what a kinetic mechanism specifies (Cleland, 1963).

Chains cannot carry stoichiometry. A chain records which items were individuated and at what cost. A process consuming two equivalents of a reactant and one consuming one produce chains differing in the number of cuts, but the framework provides no map from cut count back to coefficient: an item individuated once and used twice is indistinguishable, at the level of committed boundary, from an item individuated twice. $\square$

Remark 6.8 (The practical reading). This is why a practitioner building one ontology for a reaction domain finds that the questions will not sit together, and ends by recommending separate linked modules for reactions, substances, measurements and workflows. The module boundaries are the blocks of Theorem 6.7: standards for pathway exchange (Demir et al., 2010) and term hierarchies (Ashburner et al., 2000) sit on one side of it. We offer this as an interpretation rather than a theorem: we show the partition exists, and that its blocks correspond to a boundary practitioners draw independently.

7 The medium

The contact graph of Theorem 4.1 has a medium vertex to which everything is adjacent and which no result has yet used. Giving it content turns two annotations into computations.

Definition 7.1 (Medium as reservoir). The *medium* is a pair $\mathbf{m} = (\mu, \tau)$ where μ assigns to each chemical identity i an *ambient occupancy* $\mu(i) \geq 0$, and $\tau > 0$ is the *exchange scale*: the occupancy change the medium absorbs without itself being individuated. The weight of the edge from a leaf ℓ of identity i to $\mathbf{m}$ is

$$w(\ell, \mathbf{m}) = \beta(1 + \log(1 + \tau/\mu(i))), \quad \mu(i) > 0, \quad (10)$$

and $w(\ell, \mathbf{m}) = \infty$ when $\mu(i) = 0$.

Remark 7.2 (The constitutive choice, and what depends on it). Equation (10) is a modelling choice and we justify it by three properties rather than deriving it. It never violates the floor, since

the bracket is at least 1. Abundance is cheap to individuate against and scarcity is expensive: as $\mu \rightarrow \infty$ the weight falls to β (a leaf surrounded by copies of itself is barely distinguishable from its surroundings) and as $\mu \rightarrow 0^+$ it diverges. And only the ratio τ/μ appears, so τ and μ are never separately observable.

Every result below uses Equation (10) only through the facts that it is strictly decreasing in μ and bounded below by β . Our suite re-verifies each structural result under four unrelated weight families sharing only those two properties — logarithmic, square-root, rational and a non-smooth linear cap — so that no theorem depends on the logarithm (Figure 3(a)).

7.1 Solvent role is computed

A five-class leaf algebra with a single solvent class conflates two chemically distinct things: a small number of ordered waters that occupy defined positions, make specific hydrogen bonds and mediate recognition (Creighton, 1993; Levy and Onuchic, 2006), and the bulk remainder, which is a medium rather than a participant (Ball, 2008).

Definition 7.3 (Structural residue and role). Let ℓ be a solvent leaf of identity i , and $N(\ell)$ the leaves other than $\mathbf{m}$ adjacent to it. The *structural residue* is $\rho_{\text{str}}(\ell) = \sum_{u \in N(\ell)} w(\ell, u)$, the boundary ℓ maintains against the system, as distinct from $w(\ell, \mathbf{m})$, the boundary it maintains against its surroundings. Then

$$\text{role}(\ell) = \begin{cases} \text{structural} & \text{if } \rho_{\text{str}}(\ell) \geq w(\ell, \mathbf{m}), \\ \text{bulk} & \text{otherwise.} \end{cases} \quad (11)$$

Theorem 7.4 (Role is decidable at the floor). *For every solvent leaf, $\text{role}(\ell)$ is determined by the contact graph and the medium alone, requires no annotation, and is invariant under any rescaling of w preserving the floor. Moreover a leaf with no system contacts is **bulk**, and role is monotone in $\mu(i)$: increasing ambient occupancy can turn a bulk leaf structural but never the reverse.*

Proof. Determinacy and invariance are immediate: both sides of Equation (11) are sums of edge weights, and the comparison is a ratio test, hence invariant under $w \mapsto cw$ for $c > 0$ provided $c\beta$ is taken as the new floor. If $N(\ell) = \emptyset$ then

$\rho_{\text{str}} = 0$ while $w(\ell, \mathbf{m}) \geq \beta > 0$, so the test fails and the role is **bulk**. For monotonicity, $w(\ell, \mathbf{m})$ is strictly decreasing in $\mu(i)$ by Theorem 7.2 while ρ_{str} does not depend on it, so increasing $\mu(i)$ lowers the right-hand side and cannot falsify a passing test. $\square$

Corollary 7.5 (The role question is answered by computation). *“What is the role of the solvent in this reaction?” has an answer of the form “structural, because $\rho_{\text{str}} = x \geq w(\cdot, \mathbf{m}) = y$ ”, derived from the graph. No curator supplies it and no vocabulary term records it.*

Our suite exhibits two waters of identical chemical identity receiving opposite roles: an ordered active-site water with $\rho_{\text{str}} = 4\beta$ is **structural**, and a bulk water with no system contacts is **bulk** (Section 10.3, Figure 3(b)).

7.2 Direction is a property of the medium

A knowledge graph over reactions must record directionality and cannot do it with a boolean. The public reaction resource we queried stores 18 907 bidirectional master reactions and 37 814 directional children, exactly two per master: for the alanine transaminase reaction RHEA:19453, “L-alanine + 2-oxoglutarate = pyruvate + L-glutamate”, the children are 19454 and 19455. The same master identifier names a reaction that runs one way in one organism and the other way in another. The distinction “reversible in principle, irreversible physiologically” is real, load-bearing, and in every representation we know of it is *asserted* by a curator.

Lemma 7.6 (Reversal invariance of the chain). *Let $C = (S_1, \dots, S_k)$ be a chain with residues $\rho_i = w(\partial(S_i))$ and C^R its reversal. Then C and C^R commit the same total boundary, the same number of cut events, and the same set of committed edges.*

Proof. Each ρ_i is the weight of an edge boundary, a quantity of the graph and not of the order in which boundaries are committed. Summation is commutative, the cut count is the cardinality k , and the committed edge set is $\bigcup_i \partial(S_i)$: all three are invariant under permutation and in particular under reversal. $\square$

Corollary 7.7 (Cut structure cannot orient). *No predicate definable from the residues, the cut count, or the committed boundary distinguishes C from C^R . Direction must come from outside the chain.*

Definition 7.8 (Medium bias). For a chain from initial multiset I to terminal multiset T evaluated in medium $\mathbf{m}$, the *medium bias* is

$$\Delta_{\mathbf{m}}(C) = \sum_{i \in T} w(\ell_i, \mathbf{m}) - \sum_{i \in I} w(\ell_i, \mathbf{m}). \quad (12)$$

By Equation (10), $\Delta_{\mathbf{m}}(C) > 0$ exactly when the products are on balance *scarcer* in the medium than the reactants, and $\Delta_{\mathbf{m}}(C^R) = -\Delta_{\mathbf{m}}(C)$ by construction.

Theorem 7.9 (Direction trichotomy). *Let $\delta = \Delta_{\mathbf{m}}(C)$. Exactly one of the following holds.*

- (a) $\delta > \beta$: the process is irreversible in the direction of C .
- (b) $\delta < -\beta$: the process is irreversible in the direction of C^R .
- (c) $|\delta| \leq \beta$: the process is reversible in principle and undirected in this medium.

Consequently reversibility is a property of the pair (chain, medium) and not of the chain, and the same chain may fall in different cases in different media.

Proof. The three cases are exhaustive and mutually exclusive as a trichotomy on the real number δ against $\beta > 0$. The floor appears because a bias smaller than β is a distinction the receiver cannot make: a medium favouring one direction by less than the floor does not favour it at all. The final clause is immediate because δ depends on $\mathbf{m}$ through Equation (12) while the chain’s own admissibility does not. $\square$

Corollary 7.10 (One identifier, two directions). *Two organisms whose media differ in the ambient occupancy of a shared participant may run the same chain in opposite directions, and both are correct. The reaction identifier names the chain, which is direction-symmetric by Theorem 7.6; the direction is named by the medium, which the identifier does not carry.*

Our suite computes the trichotomy for RHEA:19453 in three media and reaches all

three cases: $\delta/\beta = +5.20$ in a glutamate-depleted cytosol (forward), -3.82 in a 2-oxoglutarate-depleted one (reverse), and exactly 0 in a balanced medium (undirected). The third is a negative control: without a medium that *refuses* to orient, the trichotomy would be a dichotomy with a decorative third branch (Figure 3(d)).

7.3 A saturation asymmetry we did not anticipate

Proposition 7.11 (Flooding cannot reverse a chain). *Let C be a balanced chain whose identities all sit at a common baseline occupancy μ_0 , and let one terminal identity be made arbitrarily abundant. Then under Equation (10)*

$$\Delta_m(C) \longrightarrow -\beta \log(1 + \tau/\mu_0) \quad \text{as } \mu \rightarrow \infty, \quad (13)$$

so the attainable bias is bounded. Depletion of a reactant carries no such bound.

Proof. Each baseline identity contributes $w = \beta(1 + \log(1 + \tau/\mu_0))$. Making one terminal identity abundant sends its contribution to $\beta(1 + \log 1) = \beta$, so the terminal sum falls by exactly $\beta \log(1 + \tau/\mu_0)$ while the initial sum is unchanged, giving Equation (13). For the second clause, $\mu \rightarrow 0^+$ sends $w(\ell, m) \rightarrow \infty$ by Equation (10), so the bias is unbounded. $\square$

Remark 7.12 (How this was found). We did not anticipate Theorem 7.11 and would not have found it by inspection. An earlier sweep of the trichotomy varied only the product-end occupancy and reached case (a) and case (c) but never case (b), which looked like a refutation of Theorem 7.9. The theorem was not at fault: the sweep was one-sided, and a one-sided sweep cannot reach case (b) for the reason now recorded as Theorem 7.11. At $\mu_0 = 10^{-4}$ and $\tau = 10^{-3}$ the measured limit is -2.397895β against the predicted $-\log(11)\beta = -2.397895\beta$, agreeing to six decimal places, and thirty orders of magnitude of further flooding move it no further; depletion over the same range reaches -22.93β and continues (Figure 3(c)).

Read chemically: *product inhibition cannot reverse a reaction; only substrate depletion can.* We report the episode because a suite written to pass would have hidden it.

Part IV

The Architecture

8 The plan language and its verdicts

8.1 Why a plan language and not a query language

Federating heterogeneous sources is an old problem with a long literature (Buil-Aranda et al., 2013; Halevy et al., 2006); the observation below is narrower. The sources a catalysis question must cross — a reaction knowledge base with a SPARQL endpoint (Bansal et al., 2022), a protein resource exposing both SPARQL and REST (The UniProt Consortium, 2023), a pathway resource exposing only flat files (Kanehisa and Goto, 2000; Milacic et al., 2024), an electronic laboratory notebook — do not share a query model, and no amount of engineering will give them one. They do share a *result* model: finite sets of identifiers carrying attributes.

Principle 8.1 (Terms denote results). A language whose terms denote result sets and whose operators denote instructions on what to do with a result is portable across backends in a way that a language whose terms denote graph patterns cannot be. Queries are its leaves and are never written by the user.

This is what removes the *querying requirement* of Section 1: the asker states what they want done with a result, not how the curator spelled it.

Definition 8.2 (Source). A *source* is a triple of an endpoint, a declared *capability set* — the structural features it can state — and an extraction function.

Theorem 8.3 (Static capability containment). *Lowering a plan step to a source is total exactly on the steps whose required capabilities are contained in the source’s declared set. Containment is decidable before any request is issued.*

Proof. A step declares a finite set r of required features; a source declares a finite set c . Lowering emits a query for each feature of r ; it succeeds exactly when each has a lowering rule, which holds exactly when $r \subseteq c$. Both sets are

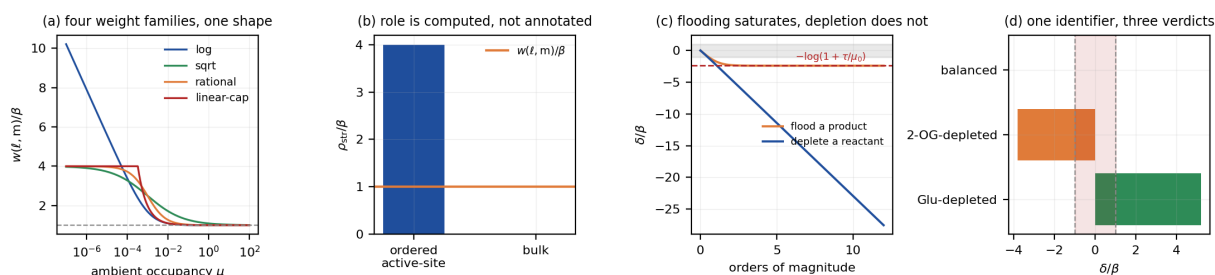


Figure 3: The medium decides solvent role and reaction direction, and one asymmetry in it was found by a test written to be capable of failing. (A) Four weight families satisfying the two properties the theorems actually use—strictly decreasing in ambient occupancy and bounded below by the floor (dashed). Logarithmic, square-root, rational and a non-smooth linear cap are otherwise unrelated. Every structural result in Section 7 is re-verified under all four, so no theorem depends on the constitutive choice of Equation (10); only the numerical values do, and those are illustrative. (B) Solvent role is computed, not annotated (Theorem 7.4). Two waters of identical chemical identity receive opposite roles: the ordered active-site water holds $\rho_{str} = 4\beta$ against the system and is **structural**; the bulk water holds none and is **bulk**. The orange line is the competing boundary $w(\ell, m) = 1.000\beta$, computed at ambient water. This is the distinction a five-class leaf algebra with one **solvent** class cannot draw, and no curator supplies it. (C) The saturation asymmetry of Proposition 7.11, which we did not anticipate. Flooding a product (orange) approaches the ceiling $-\log(1 + \tau/\mu_0) = -2.397895\beta$ (red dashed) and stops: thirty orders of magnitude of further flooding move it no further, and the measured limit agrees with the predicted one to six decimal places. Depleting a reactant (blue) descends without bound, reaching -22.93β over the same range. Read chemically, product inhibition cannot reverse a reaction and only substrate depletion can. The grey band is the refusal region $|\delta| \leq \beta$. This result exists because an earlier one-sided sweep never reached case (b) of Theorem 7.9 and looked like a refutation; the theorem was not at fault, the sweep was. (D) One reaction identifier, three verdicts (Corollary 7.10). The same chain, with the same participants and the same identifier, is forward at $\delta = +5.204\beta$ in a glutamate-depleted cytosol, reverse at -3.818β in a 2-oxoglutarate-depleted one, and undirected at exactly 0 in a balanced medium. The third is a negative control: without a medium that *refuses* to orient, the trichotomy would be a dichotomy with a decorative third branch.

finite and given in advance, so $r \subseteq c$ is decidable statically. $\square$

Remark 8.4 (Capability is a claim about a compiler). “This system supports counting” is true or false of a lowering procedure, not of the target logic. The two come apart in both directions: a formalism may express what no compiler lowers, and a compiler may lower what the formalism does not entail. Only the compiler-level reading admits the decidable check of Theorem 8.3.

8.2 Six verdicts

Definition 8.5 (Verdict). A step returns one of

answer	a result set, certified non-empty
empty	the question is well posed and the answer is certified empty
unexpressed	the question cannot be stated in the source’s model
unsupported	statable in the model, not lowerable by this compiler
starved	this step failed because an earlier step under-retrieved
exhausted	the budget ran out before the step completed

Theorem 8.6 (Non-degeneracy). *No verdict other than **answer** may carry a non-empty payload, and **answer** may not carry an empty one unless emptiness is itself certified.*

Proof. Suppose a verdict $V \neq \text{answer}$ carried a non-empty payload. A caller consuming payloads would then act on results from a step reporting that it did not answer, so V and **answer** would be indistinguishable to that caller, collapsing the verdict space to the boolean it exists to refine. The second clause is symmetric: an uncertified empty **answer** is exactly the conflation of Section 1.2. $\square$

This is a structural property, not a reporting discipline. In our reference kernel it is enforced by the constructor: attempting to build a **starved** result carrying a payload raises, and our suite checks that it does (Section 10.3).

Remark 8.7 (On budgets). A plan spends a finite retrieval budget across sources whose yields differ. Where each source’s yield is concave in the effort spent on it, the optimal split equalises marginal return — the water-filling rule familiar from resource allocation under a concave objective (Boyd and Vandenberghe, 2004). We do not develop the allocator here, and note only that a budget exceeding the sum of minimum step costs is necessary but not sufficient for a

plan to complete, since a step may starve on a predecessor rather than on its own funding. Enumeration over a handful of federated sources is affordable (Cormen et al., 2009).

Proposition 8.8 (Starvation is the characteristic federated failure). *starved is reachable only when a step’s input is produced by another step, which is the federated setting. A starved step must name the predecessor at fault, and a walk up those names terminates.*

Proof. In a single-source query a step’s input is the corpus, which does not under-retrieve; the verdict is unreachable. In a plan, step j may receive empty bindings from step $i < j$ although i answered correctly under its own declaration. Blame is assigned to i ; since indices strictly decrease, the walk terminates. $\square$

8.3 The stencil

We now state what the architecture does with a fetched record.

Construction 8.9 (Ladder as stencil). Given records fetched by a plan, translate them into a contact graph (Theorem 4.1), declare a medium (Theorem 7.1), and lay a ladder over the result: each contact in the process contributes a rung whose power is computed from Equation (6) using the separation costs of the translated graph. The composite power, the invariants of Theorem 5.12, the role of Equation (11) and the direction of Theorem 7.9 are then read off.

Remark 8.10 (Why this removes the modelling requirement). The stencil asks of a record only what is needed to compute a separation cost. It does not ask what class the record belongs to, which ontology term names its relation, or whether a curator anticipated the question. A question such as “in which direction is this reaction admissible here” is therefore answerable over records whose schema contains no notion of direction — which is the sense in which one may query items one cannot model.

Remark 8.11 (What the stencil costs, stated plainly). Two costs are real and we name them. First, the translation from a record to a contact graph supplies structure the record does not state — implicit hydrogens, inferred bond orders, a medium the record never mentions — and a downstream reader cannot distinguish a

```

plan bacterial_transaminase_no_cys

  source RXN capability { reaction,
    participant, ec, equation }
  source PROT capability { protein, organism,
    lineage, sequence }

  let rxns = fetch RXN where participant
    contains "benzylethylamine"
  let enzs = fetch PROT where catalyses in
    rxns
  let bact = keep enzs where lineage contains
    "Bacteria"
  let final = keep bact where not contains(
    sequence, "C")

  require final nonempty
  emit final

```

Figure 4: A plan for question Q1. The user states what is wanted of a result; no SPARQL is written. The last step is a predicate over a retrieved attribute and is not a triple lookup: no public store holds a `hasCysteine` flag, so a pattern-matching query returns nothing (Section 10.2). The plan is checked against the declared capabilities before any request is issued (Theorem 8.3).

stated element from a supplied one unless the translator tags them. Second, the powers used in Section 10.3 are *declared* rather than derived from measured kinetics. Deriving them from retrieved data is the natural next step and we have not taken it; Section 11 records both.

Part V Validation

9 Protocol

The suite is designed on one principle: *every check must be able to fail*. Four experiments run in dependency order. Experiments A–C reach no network and resolve against fixtures, so every number is reproducible offline. Experiment D reaches a public endpoint and reports the live arm as untested rather than failed if the service is unreachable.

Three devices make the suite capable of failing.

1. **Negative controls.** Four checks assert that something does *not* happen on well-formed

input: that a constant statistic does not discriminate; that the query battery of Section 10.2 separates a genuinely different contact relation; that a balanced medium refuses to orient; and that the miniature of Section 10.4 returns nothing when asked for a root present in no record.

2. **Independent cross-checks.** The SPARQL results are computed on two independently developed engines, and the analytic sensitivity is checked against a numerical derivative that does not use the closed form.
3. **Non-discriminating checks are excluded from the score.** One check — permuting a fixed multiset of powers — cannot fail, because Equation (7) is symmetric in its arguments. The hazard of a check that is present but does no work is the analogue of vacuous satisfaction in model checking (Beer et al., 2001; Kupferman and Vardi, 2003). We report it as non-discriminating and exclude it rather than counting a pass.

Software: Python 3.14, `rdflib` 7.6.0 (RDFLib Team, 2026), `pyoxigraph` 0.5.9 (Pellissier Tanon, 2026), NumPy 2.4.3 (Harris et al., 2020). All measurements are dated 3 September 2026.

10 Results

10.1 Experiment A: the ladder algebra

Twenty-three scored checks; all pass. One further check is excluded as non-discriminating, and one is a negative control.

The floor is computed rather than assumed on three instances, including an adversarial one whose smallest edge weight is 10^{-12} ; every instance returns a strictly positive floor. On that instance the floor is $1.0 + 10^{-12}$ and *not* 10^{-12} , confirming Theorem 4.5: a cheap edge inside an expensive cut does not make the cut cheap. The complementary branch is exhibited separately — a system refining without limit produces thicknesses $1/n$ whose infimum is zero, and truncating to ten stages restores a floor at 0.1 — so the hypothesis of Theorem 4.6 does real work.

Theorem 4.7 is measured: in a graph with a four-vertex core of internal weight 50 joined to the periphery by a single edge, the minimiser

for a core vertex contains all four core vertices at cost 1, and no singleton attains it.

All four composition laws are scored against rung-by-rung simulation over 4000 ladders. The multiplicative law is exact to machine precision (0.0); the additive, maximum and mean laws err by 5.45×10^{-2} , 1.49×10^{-1} and 4.31×10^{-1} . Scoring only the favoured law would not have distinguished it.

The sensitivity correction of Theorem 5.11 is measured on 5000 ladders: the additive argmax is the highest-power rung in 5000/5000 cases, and the proportional spread across rungs is 1.11×10^{-16} in 5000/5000. The analytic derivative agrees with a numerical one to 1.3×10^{-10} .

For closed ladders, ϱ reparametrises composite power on linear ladders to 1.4×10^{-13} over 500 trials — the redundancy of Theorem 5.14 made quantitative — while v separates 2518 of 4000 ladder pairs constructed to share a composite power exactly.

10.2 Experiment B: expressibility

Nine scored checks pass, one of them a negative control.

Two contact graphs are constructed as in Theorem 6.4. They record identical triples: six each, set difference empty. Their separation costs agree, $\sigma(v_0, x) = 2$ in both, and their floors differ, $\beta_1 = 1$ and $\beta_2 = 2$, so the accountability verdicts are opposite — inadmissible in the first and admissible in the second.

Thirteen retrieval query forms were then run against both graphs on both engines: all triples, triple count, per-node degree, neighbours, two ASK forms over property paths, transitive closure, two-step neighbourhoods, adjacency to the medium, distinct subject count, a two-hop join, an OPTIONAL with FILTER, and a nested aggregate over grouped degree. **No query separated the two graphs**, and the two engines agreed on every query, 13/13.

The negative control is the load-bearing row. The same battery run against a third graph with a genuinely different contact relation separates it on 8 of 13 queries. Without this, the result above would be consistent with a blind battery.

Theorem 6.6 is executed: with the weights stored as triple attributes, a query for the minimum stored weight returns 1.0 for *both* graphs, while their floors are 1 and 2. Storing the num-

bers does not make the minimum-over-subsets expressible.

Finally the Q1 shape is executed. A query for a `hasCysteine` flag returns nothing, because no such predicate was curated; the same question asked as a computation over the retrieved sequence returns the correct two proteins.

10.3 Experiment C: twenty-eight questions

Twenty scored checks pass, one of them a negative control.

Twelve questions were supplied by practitioners — five from biocatalysis and seven generic dataset queries in the shape of a chemistry data catalogue profile — without reference to this framework. We added sixteen to exercise verdicts the twelve do not reach. Table 1 gives the resolution of all twenty-eight. Twenty-one are answered and seven refused, and five distinct verdicts occur.

Four findings are worth stating beyond the pass count.

The dataset filters discriminate. Questions Q6–Q8 form a strictly narrowing chain: three datasets mention the compound, two of those are of the requested activity type, and one of those was measured on the named vendor’s instrument *at the named setting*. Each added constraint removes at least one dataset, so the filters are shown to discriminate rather than merely to return a row. Restricting from “a Bruker” to “a Bruker set to ^{13}C ” removes a dataset: a configuration value is part of a measurement’s provenance.

One identifier, three verdicts. The direction trichotomy reaches all three cases on the alanine transaminase chain, as reported in Section 7. The balanced medium is a negative control and returns $\delta = 0$ exactly, refusing to orient.

Two waters, opposite roles. Theorem 7.4 is measured with $w(\ell, \mathbf{m}) = 3.700 \times 10^{-4}$: the ordered active-site water has $\rho_{\text{str}} = 1.480 \times 10^{-3} \geq w$ and is **structural**; the bulk water has $\rho_{\text{str}} = 0 < w$ and is **bulk**.

The refusals carry remedies. All seven refusals name both a blocker and an unblock path, and the four verdict kinds reached are distinguished by different mechanisms: **unexpressed** by static capability containment, **unsupported** by a question that is not of retrieval type, **starved** by a predecessor that under-retrieved, and **exhausted** by a budget. In the budget case two steps answer and two report **exhausted**; a rows-only interface would have returned the same empty table for both.

10.4 Experiment D: route divergence

Five scored checks pass, one of them a negative control.

Two SPARQL queries were run against a public biochemical reaction endpoint on 3 September 2026. They are byte-identical except for one line: Form A binds two variables through an object list, `?r p1/p2/p3/p4 ?a , ?o`, and Form B through two separate triple patterns.

Form	Distinguishing line
A (object list)	<code>... rh:chebi ?a , ?o .</code>
B (two patterns)	<code>... rh:chebi ?a rh:chebi ?o .</code>

The elapsed times are reported because they are the one piece of diagnostic evidence the measurement carries. Form A returns a smaller answer roughly fourteen times faster, which is the signature of a route that explored less of the store rather than of one that explored the same store and found less. We draw no conclusion from it — timing against a shared public service is confounded by load, caching and concurrent queries — but a reader reconstructing the observation should have the numbers.

The two must denote the same query. SPARQL 1.1 defines an object list as an abbreviation: `?s ?p ?o1 , ?o2` expands to `?s ?p ?o1 . ?s ?p ?o2 .`, and the expansion is performed during translation to the abstract query, before any evaluation semantics is reached (Har-[ris and Seaborne, 2013](#)). Nothing in the expansion or in the property-path semantics makes it conditional on the predicate being a bare IRI.

The local control settles what the semantics gives. Both spellings were run unmodified against a hand-checkable miniature of the same data shape — two reactions, one qualifying —

Table 1: All twenty-eight questions and their verdicts. Q1–Q12 were supplied by practitioners; Q4' and Q13–Q28 were added to exercise outcomes the twelve do not reach. Every refusal carries a blocker and an unblock path; those are abbreviated here and given in full in the results file.

#	Question	Verdict	B
<i>Group 1 — biocatalysis (practitioner-supplied)</i>			
Q1	bacterial transaminase, no Cys	answer	2
Q2	buffer and pH for mt-X	answer	la
Q3	substrate scope of BVMO-Y	answer	ag
Q4	expected products, kinetic resolution	unsupported	no
Q5	device and wavelength, 23 Mar	answer	in
<i>Group 2 — generic dataset queries</i>			
Q6	datasets about compound C	answer	3
Q7	... of activity type T	answer	na
Q8	... on a Bruker set to X	answer	na
Q9	datasets about a reaction with product P	answer	4
Q10	... with starting material C	answer	4
Q11	... measured on 2UV-vis with C	answer	1
Q12	... with catalyst S on 3Bruker	answer	3
<i>Group 3 — answerable without a model of the answer</i>			
Q4'	direction of RXN 19453 per medium	answer	th
Q13	role of the solvent	answer	co
Q14	which catalysts are interchangeable	answer	on
Q15	which step may be deleted	answer	pr
<i>Group 4 — refused, with named blockers</i>			
Q16	derived reaction temperature	unexpressed	fl
Q17	buffer, from the reaction store	unexpressed	ca
Q18	enzymes for an unretrieved reaction	starved	pr
Q19	four-step plan on a budget of two	exhausted	tw
<i>Group 5 — extended battery</i>			
Q21	reactions running both ways	answer	tw
Q22	bacterial enzymes for an EC	answer	2
Q23	pathways producing a lactone	answer	1
Q24	experiments above pH 8	answer	1
Q25	operators using UV-vis	answer	1
Q26	datasets from a ¹³ C setting	answer	2
Q27	which enzyme gives highest <i>ee</i>	unsupported	no
Q28	ordered steps within one turnover	unexpressed	si

on `rdflib` and `pyoxigraph`. All four runs return the qualifying reaction and agree with hand computation. A further control confirms the miniature can return nothing when asked for a root present in no record, so the agreement is not an artefact of a query matching everything.

Observation 10.1 (Status of the finding). **Divergence: established.** Two spellings the specification declares equivalent, verified equivalent on two independent engines against a hand-checked miniature, return 2 and 397 against the same endpoint on the same date. **Cause: not established.** Query planning, an internal traversal cap, and evaluation-order effects on the shared property path all remain live candidates. We did not investigate further, because doing so would mean probing a third party’s service.

Remark 10.2 (A confound, reported against our own interest). The endpoint’s loaded ontology carried 217 544 classes on the date of measurement. Any count comparison between the endpoint and a separately published download of the same ontology is therefore confounded: the two are different snapshots. An investigator attempting to adjudicate the pair (2, 397) by reproducing the query against a download is not running a control; they are running a third route.

10.5 Reading divergence as coverage

The natural conclusion from Theorem 10.1 is that something is broken. A different reading is available and, in our judgement, more useful.

Theorem 10.3 (Routes cannot contradict). *Let two routes share endpoints and converge to the same fixed point under a propagation model of retrieval. Then neither returns an item the other excludes: their answers are nested, and the numerical difference measures how much of the underlying network each resolved.*

Proof. Membership in the answer is defined by *existence* of a convergent sequence from the query to the item, not by choice of one. If both routes converge to the same fixed point, an item exhibited by one is reachable in the other, so the answers cannot conflict; they may differ only in how much of the reachable set each route actually enumerated. $\square$

On that reading the pair (2, 397) is a measurement of resolved extent, and reporting it as “one

query is broken” discards the quantity it carries. We are explicit that the reclassification is *conditional*: it requires both routes to converge to the same fixed point, which for this endpoint we could not verify from outside. Without that check the ordinary defect reading stands.

The practical consequence costs nothing to adopt.

- (P1) Record both answers; keeping both costs one integer.
- (P2) Report the difference as coverage, not as error. “Routes A and B differ by 395” is more informative than “route A failed”.
- (P3) Check the convergence hypothesis before reclassifying, or a genuine fault will be absorbed as a coverage difference.
- (P4) Keep at least one route strict enough that its disagreement is meaningful.
- (P5) Never compare counts across snapshots. This one is unconditional.

10.6 What running the suite changed

Four episodes altered what we believed, and each is load-bearing enough to state rather than to absorb silently.

A hand-computed floor that was wrong, in the way the paper warns about. Our first expectation for the small graph of Section 10.1 was $\beta = 2$, taken from the lightest edge incident to a vertex. The computed value is 5, attained by the *region* $\{a, b\}$ rather than by either vertex alone: separating a costs $\min(2 + 5, 2 + 3) = 5$. The error is exactly the one Theorem 4.5 and Theorem 4.7 exist to forbid — reading a separation cost as an edge weight — and we had made it while writing the section that forbids it. The check now records the expected value as a separation cost and carries the reasoning in a comment.

An adversarial instance that does not do what its name suggests. We included a graph whose smallest edge is 10^{-12} expecting the floor to follow it down. It does not: the floor is $1 + 10^{-12}$, because that tiny edge lies inside every cut that would use it, and a cheap edge inside an expensive cut does not make the cut cheap.

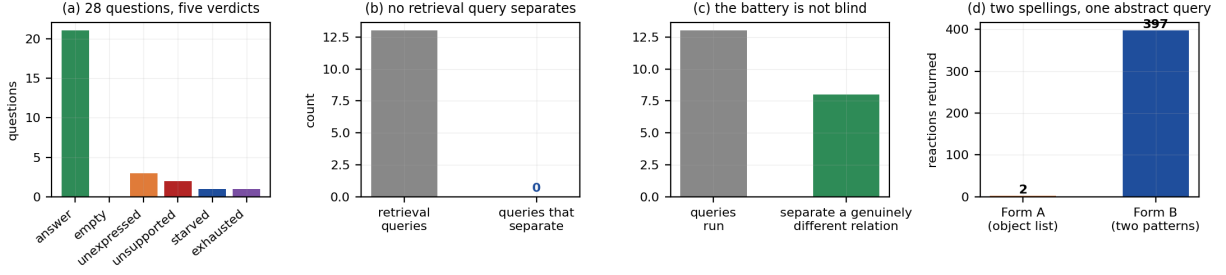


Figure 5: Twenty-eight questions, five verdicts, and the expressibility separation executed rather than argued. (A) The twenty-eight questions by verdict: 21 answered, 3 unexpressed, 2 unsupported, 1 starved and 1 exhausted. Twelve of the questions were supplied by practitioners without reference to this framework and sixteen were added to reach outcomes the twelve do not. Every refusal carries a named blocker and a statement of what would unblock it. A system that answered all twenty-eight would have an empty defined class: it would not classify, it would merely accept. (B) The separation of Theorem 6.4, measured. Thirteen retrieval query forms—triple enumeration, counts, grouped degree, neighbourhoods, two ASK forms over property paths, transitive closure, two-hop joins, OPTIONAL with FILTER, and a nested aggregate—were run against two contact graphs recording identical triples, on two independently developed SPARQL engines (rdflib 7.6.0 and pyoxigraph 0.5.9). The two engines agreed on every query, and none of the thirteen separated the two systems, while their accountability verdicts differ: $\sigma(v_0, x) = 2$ in both, with floors 1 and 2. (C) The negative control, which is the load-bearing panel. The same battery run against a third graph with a genuinely different contact relation separates it on 8 of the 13 queries. Without this, the null result in (B) would be consistent with a blind battery rather than with a property of the two systems. (D) The live divergence of Section 10.4. Two spellings that SPARQL 1.1 defines as one abstract query—an object list against two triple patterns—return 2 and 397 against the same public endpoint on the same date, while both spellings agree with each other and with hand computation on a miniature of the same data shape run on both local engines. The divergence is established; its cause is not, and we declined to diagnose a third party’s service.

The instance is retained, with its expectation corrected, because it is a sharper illustration of Theorem 4.5 than the one we designed.

A one-sided sweep that looked like a refutation. Theorem 7.12 records this in full. The short form: a sweep varying only the product-end occupancy reached cases (a) and (c) of Theorem 7.9 and never case (b), which presents as a theorem with an unreachable branch. The theorem was not at fault; the sweep was, and repairing it produced Theorem 7.11, which we had not anticipated and would not have found by inspection.

A check that was scored when it should have been excluded. The permutation check of Section 9 was initially counted among the passes. It cannot fail — Equation (7) is symmetric in its arguments — so counting it inflated the score by one and told the reader nothing. It is now excluded and reported as excluded. We mention it because the difference between 58 checks and 57 scored checks is precisely the difference between a suite that reports what it

measured and one that reports how many boxes were green.

Remark 10.4 (Two figures that would have hidden a result). Two panels were redrawn after they were first plotted. Figure 3(c) originally showed a single depletion sweep on axes that made the flooding branch look like an unbounded descent; the ceiling of Theorem 7.11 was off-scale and invisible. Figure 2(a) was first drawn without a legend distinguishing the four composition laws, which made the comparison — the whole content of the panel — unreadable. In both cases the figure was drawn from correct numbers and still conveyed the wrong thing, which is a reminder that a computed panel is not automatically an honest one.

10.7 Summary of outcomes

Table 2: Validation summary. Of 58 reported checks, 57 are scored and all pass; 4 of those are negative controls that must fail on well-formed input and do, and 1 further check is recorded non-discriminating and excluded from the score.

Exp.	Claim checked	Scored
A	Theorems 4.4, 5.6, 5.9 and 5.10	23
B	Theorems 6.4 and 6.6	9
C	Theorems 7.4, 7.9, 7.11 and 8.6	20
D	Theorem 10.1, the local control	5
total		57

We state this explicitly because a table of successes is not by itself informative. What makes the suite meaningful is that each check was constructed so that it could have failed, that three are constructed so that they must fail, and that one was found unable to discriminate and was removed from the score rather than counted.

Part VI Scope

11 What this does not do

We state the limits plainly, because several are severe enough that a reader could otherwise take the paper to claim more than it does.

No experimental validation. Nothing here has been compared against a measured hydration structure, a measured direction reversal, a measured ambient occupancy, or a measured rate. Theorem 7.9 is worked from the qualitative fact that organisms differ in metabolite demand, not from measured intracellular concentrations. Theorem 7.10 is an explanation of a documented annotation practice, not a prediction tested against one.

The powers are declared, not derived. In Section 10.3 and Table 1 the rung powers are literals. Deriving them from retrieved data — and answering what a derived power means for a result set — is the substantive next step and is not taken here. Until it is, the ladder demonstrates a *form* of answer on catalysis data rather than

a measured one; measured enzyme parameters cluster well away from physical limits (Bar-Even et al., 2011), and reproducing that distribution would be a real test.

Translation supplies structure. Converting a record to a contact graph adds implicit hydrogens, inferred bond orders and a medium the record never mentions. Unless the translator tags each element as stated or supplied, a downstream reader cannot tell a recorded fact from a convention. We regard provenance tagging as a requirement of any serious implementation and have not implemented it here.

The medium weight is a modelling choice. Equation (10) is justified by three properties and by reduction to a chemical potential in the dilute limit, but it is not derived from Theorem 4.4. A different monotone, floor-bounded function gives the same theorems with different constants, which is why the suite re-runs every structural result under four families. Every numerical statement about the medium is therefore illustrative; the structural statements are not.

No kinetics and no thermodynamics. The medium bias is not a free energy: it is a difference of individuation costs. Connecting Δ_m to a reaction free energy (Beard and Qian, 2007; Noor et al., 2013) would be a substantial and falsifiable next step and is not taken. In consequence “at what temperature does this reaction take place” remains open: the floor is condition-dependent in principle, but at the categorical depth used here the dependence is immaterial, and we would rather say so than report a number a laboratory cannot use.

Fixtures are not the public resources. Experiments A–C run against small hand-built stand-ins with the same record shape as the public sources, not against copies of them. This makes every number reproducible offline and makes none of them evidence about biology. Only Experiment D touches a live service.

Capability declarations are asserted. A source’s capability set is declared by its adapter and is not verified against the service. A wrong declaration produces a wrong static verdict, and

Theorem 8.3 guarantees only that the check is decidable, not that the input is true.

One construct is not a general claim. The ladder is local, pure and cheap. A construct that had to issue its own requests, or to fail in a way the six verdicts do not describe, would test the architecture far harder, and nothing here predicts how that would go.

12 Conclusion

A knowledge graph is only as useful as the person who built it and the person querying it. Both requirements bind, and we have shown an architecture in which neither binds for an important class of question.

The reason it works is a theorem rather than an engineering trick. An answer set is a substructure of what was stored, so a question whose answer depends on a quantity nobody stored has no expression in a retrieval language — however good the ontology. We exhibited two systems with identical triples whose accountability verdicts differ, ran thirteen retrieval query forms against them on two independent engines without separating them, and confirmed with a control that the battery is not blind. Storing the weights does not help, because the quantity that decides is a minimum over subsets and no basic graph pattern forms one.

What replaces the stored answer is a computation over fetched records. Records are gathered by a plan whose terms denote results rather than graph patterns, translated into contact graphs, and stencilled with a ladder. Because the stencil asks only for a separation cost, it applies to records whose schema has no notion of the thing being asked — which is how one queries items one cannot model. The solvent’s role becomes a comparison of two boundaries rather than a curator’s annotation; a reaction’s direction becomes a trichotomy on one inequality, so that one identifier can correctly name a process running opposite ways in two organisms.

The framework refuses. Seven of twenty-eight questions are declined, each naming a blocker and a remedy, and refusal is enforced structurally: no verdict but `answer` may carry a payload. A framework that classifies everything has an empty defined class.

Two results came from tests written to be capable of failing. A one-sided sweep of the direction trichotomy never reached its third case, and the reason turned out to be a saturation bound we had not anticipated: flooding a product cannot reverse a chain, because the attainable bias stops at $-\log(1 + \tau/\mu_0)$, while depleting a reactant is unbounded. And the counter-intuitive claim that control lies at the strongest rung survives only under an additive parametrisation; under proportional improvement the sensitivity is flat, at machine zero over five thousand ladders. We report both because a suite written to pass would have hidden them.

What remains is the measurement. The powers here are declared and not derived from kinetics, no number has been checked against a laboratory, and the translation from record to graph supplies more than it states. Those are the next experiments, and they are the ones that would make the architecture a scientific instrument rather than a well-founded proposal.

Reproducibility

All experiments are in <https://github.com/fullscreen-triangle/levinthal/tr>. Running `python exp_a_ladder.py`, `exp_b_expressibility.py`, `exp_c_questions.py` and `exp_d_divergence.py` reproduces every number quoted, writing per-experiment JSON to `results/`; `make_figures.py` regenerates every panel from those files. The reference kernel is `kernel/ladder.py` and `kernel/plan.py`; the fixtures are `fixtures/corpus.py`. Experiments A–C require no network. Experiment D contacts a public SPARQL endpoint and degrades to its local control if the service is unavailable.

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